

CB2207 - Structural Bioinformatics and Drug Discovery

(Project – 15%)

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Abstract:

Allergic adenoid hypertrophy is associated with chronic inflammatory signaling involving multiple cytokines, including interleukin-6 (IL6), interleukin-4 (IL4), and interleukin-13 (IL13).

Persistent cytokine activation contributes to tissue inflammation, airway obstruction, and prolonged immune responses. RNA interference (RNAi)-based therapeutics provide a potential strategy for selective suppression of inflammatory mediators.

This study aimed to computationally design and evaluate small interfering RNA (siRNA) candidates targeting IL6 while integrating biological context from the IL4–IL13 allergic inflammatory axis.

The workflow involved transcript retrieval, siRNA candidate generation, RNA secondary structure prediction, duplex interaction analysis, thermodynamic evaluation, specificity screening, biological interpretation, and conceptual delivery strategies.

IL6 transcript sequences were retrieved from NCBI and candidate siRNAs were generated using sequence-based design principles. RNA secondary structure prediction using ViennaRNA identified accessible regions potentially favorable for siRNA binding. Duplex interaction analysis suggested stable siRNA–mRNA interactions with favorable binding energies. Thermodynamic scoring integrated accessibility and asymmetry features to prioritize candidates. BLAST-based off-target analysis identified potential transcript similarities requiring further specificity evaluation.

The findings indicate that several IL6-targeting siRNA candidates demonstrate promising computational characteristics including favorable accessibility, thermodynamic stability, and predicted interaction strength. However, additional experimental validation and advanced simulations remain necessary before therapeutic translation.

The study highlights how computational bioinformatics workflows may support early-stage development of RNA-based therapeutics targeting inflammatory pathways.

Introduction:

Adenoid hypertrophy refers to enlargement of adenoid tissue located within the nasopharyngeal region and is commonly associated with recurrent infections, allergic responses, and chronic inflammation. In pediatric populations, persistent adenoid enlargement may contribute to airway obstruction, snoring, recurrent infections, and impaired quality of life.

Inflammatory signaling plays a central role in progression of allergic airway diseases. Cytokines regulate immune responses through complex signaling networks and prolonged dysregulation may lead to chronic inflammatory conditions.

Among these cytokines, **Interleukin-6 (IL6)** functions as an important mediator involved in:

- inflammatory amplification
- immune cell recruitment
- T-cell differentiation

- B-cell activation
- chronic inflammatory progression

Elevated IL6 expression has been associated with prolonged inflammatory states and tissue pathology.

Simultaneously, allergic inflammation frequently involves activation of **IL4 and IL13**, two cytokines strongly associated with Th2 immune responses. These molecules contribute to:

- IgE production
- eosinophilic inflammation
- mucus secretion
- airway remodeling
- persistent allergic responses

Interactions among IL4, IL13, and IL6 may collectively influence inflammatory severity.

Recent advances in RNA therapeutics have increased interest in **small interfering RNA (siRNA)** as targeted treatment strategies. siRNA molecules induce post-transcriptional gene silencing through RNA interference mechanisms, reducing expression of selected genes.

Compared with traditional inhibitors, siRNA-based approaches may provide:

- greater target specificity
- customizable sequence design
- reduced off-target protein interactions
- potential modulation of disease-associated pathways

However, challenges remain, including:

- off-target effects
- sequence specificity
- RNA stability
- delivery efficiency
- systemic immune consequences

Computational bioinformatics tools enable early-stage evaluation of these factors before experimental validation. Therefore, this study explores computational design and evaluation of IL6-targeting siRNA candidates while considering broader cytokine interactions and potential therapeutic implications.

Problem Statement:

Allergic adenoid hypertrophy involves chronic inflammatory signaling mediated by interconnected cytokine networks.

Although IL6 contributes to inflammatory amplification, selective suppression of IL6 without affecting broader immune balance remains challenging.

Conventional therapeutic approaches may fail to provide targeted regulation of cytokine expression and may produce systemic effects.

Therefore there is a need to computationally evaluate sequence-specific RNA therapeutics capable of selectively inhibiting IL6 while minimizing off-target interactions and considering broader cytokine pathways.

Objectives:

The present study aimed to:

Primary objective

Design and computationally evaluate siRNA molecules targeting IL6.

Specific objectives

1. Retrieve and characterize IL6 transcript sequences.
2. Generate candidate siRNA molecules.
3. Predict IL6 mRNA secondary structure.

- Identify accessible regions suitable for siRNA binding.
- Evaluate duplex formation and interaction energy.
- Assess thermodynamic favorability.
- Screen for potential off-target interactions.
- Interpret findings within IL4–IL13 inflammatory pathways.
- Explore conceptual delivery strategies to reduce systemic effects.

Hypothesis:

siRNA candidates targeting structurally accessible regions of IL6 mRNA and exhibiting favorable thermodynamic properties will demonstrate stronger predicted interaction stability and reduced off-target risk. Such candidates may represent potential therapeutic approaches for reducing inflammatory amplification associated with allergic disease.

Methodology:

The study followed a computational pipeline for identification and evaluation of IL6-targeting siRNA candidates. The workflow integrated transcript retrieval, sequence design, structural analysis, thermodynamic evaluation, specificity screening, and biological interpretation.

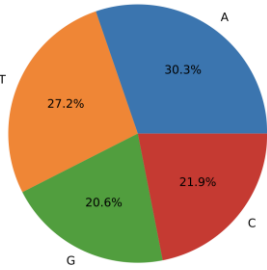
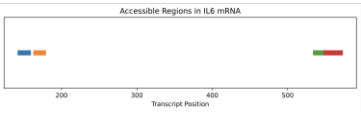
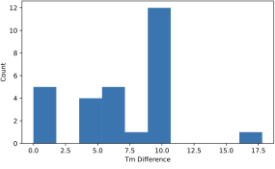
The overall workflow consisted of:

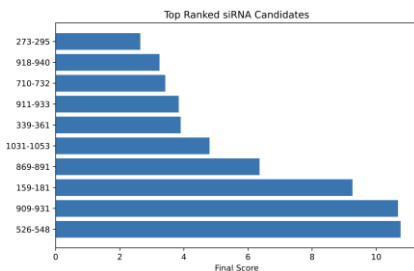
- Target selection and transcript retrieval
- siRNA candidate generation
- RNA secondary structure prediction
- siRNA–mRNA duplex interaction analysis
- Thermodynamic and stability assessment
- Off-target analysis
- Biological context integration
- Stability interpretation
- Conceptual therapeutic delivery strategy

Step	Objective	Tool(s) Used	Justification for Tool Selection	Output Generated	Interpretation of Results
Step 1 Target Selection	Retrieve canonical IL6 transcript and characterize sequence properties	NCBI Entrez, Biopython	NCBI provides validated transcript databases; Biopython enables reproducible sequence retrieval and automated analysis	IL6_mRNA.fasta, transcript_info.csv	Canonical IL6 transcript (NM_000600.5) selected as therapeutic target; transcript properties used for downstream design
Step 2 siRNA Design	Generate candidate siRNAs targeting IL6	siDirect, Python filtering	siDirect incorporates established design rules including GC optimization, asymmetry, and specificity considerations	siRNA_candidates.csv, ranked_siRNA.csv	Multiple candidates generated with varying thermodynamic properties and predicted efficacy
Step 3 RNA Secondary Structure Prediction	Identify accessible mRNA regions suitable for siRNA binding	ViennaRNA RNAfold	RNAfold predicts secondary structure using minimum free energy models; accessible loops improve siRNA binding probability	accessible_regions.csv, IL6_structure.txt	Accessible regions identified, indicating potential favorable target sites for silencing
Step 4 siRNA–mRNA Interaction Analysis	Evaluate duplex formation and interaction strength	RNA duplex (ViennaRNA)	RNA duplex estimates hybridization energy and predicts interaction stability between RNA molecules	duplex_results.csv, duplex_structure.txt	Strong duplex formation and binding energy (~-33 kcal/mol) suggested stable target interaction

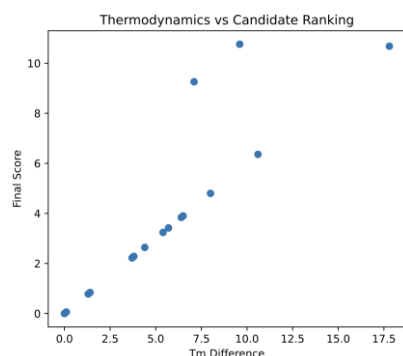
Step 5 Thermodynamic & Stability Analysis	Rank candidates based on energetic favorability	Python, Pandas custom scoring framework	Allowed integration of multiple parameters (Tm difference + accessibility + ranking) not captured by a single tool	final_ranked_siRNA.csv	Candidates with favorable accessibility and thermodynamic asymmetry received higher scores
Step 6 Off-target Analysis	Assess specificity and unintended transcript interactions	NCBI BLAST+	Standard tool for sequence similarity screening and specificity evaluation	blast_hits.csv, blast_hits.txt	Some transcript similarities detected; off-target effects cannot be completely excluded
Step 7 Biological Context Integration	Interpret findings within cytokine networks	Literature review and pathway interpretation	Computational results require biological context to evaluate therapeutic relevance	Integrated discussion section	IL6 suppression may reduce inflammatory amplification but broader cytokine balance must be considered
Step 8 Stability Evaluation	Approximate persistence and stability of selected candidates	Energetic interpretation, thermodynamic analysis	Full MD simulation required validated 3D structures and computational resources beyond project scope	md_stability.csv	Favorable energetic properties suggested stable candidate behavior
Step 9 Conceptual Targeting Strategy	Explore localized delivery approaches	Literature review, translational interpretation	Therapeutic efficacy depends on delivery efficiency and reduction of systemic effects	Conceptual targeting framework	Intranasal delivery and nanoparticle systems may improve specificity while reducing systemic cytokine

Results & Discussion:

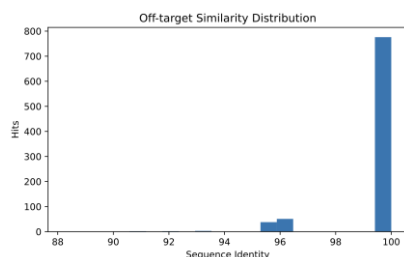
Figure	Interpretation and Discussion																		
IL6 Transcript Nucleotide Composition  <table border="1"> <caption>IL6 Transcript Nucleotide Composition Data</caption> <thead> <tr> <th>Nucleotide</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>30.3%</td> </tr> <tr> <td>T</td> <td>27.2%</td> </tr> <tr> <td>G</td> <td>20.6%</td> </tr> <tr> <td>C</td> <td>21.9%</td> </tr> </tbody> </table>	Nucleotide	Percentage	A	30.3%	T	27.2%	G	20.6%	C	21.9%	The nucleotide composition analysis showed relative distribution of A, T, G, and C bases within the selected IL6 transcript. Transcript composition influences RNA stability and secondary structure formation. Moderate GC representation suggests balanced structural stability without excessive rigidity, supporting suitability for downstream siRNA targeting.								
Nucleotide	Percentage																		
A	30.3%																		
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Accessible Regions in IL6 mRNA 	Accessible regions predicted using RNAfold correspond to loops and unpaired transcript segments where siRNA binding is more likely. These regions are important because structurally inaccessible sites may reduce silencing efficiency despite sequence complementarity. Accessible regions identified in this study were incorporated into candidate prioritization.																		
Distribution of Thermodynamic Asymmetry  <table border="1"> <caption>Distribution of Thermodynamic Asymmetry Data</caption> <thead> <tr> <th>Tm Difference</th> <th>Count</th> </tr> </thead> <tbody> <tr> <td>0.0</td> <td>5</td> </tr> <tr> <td>2.5</td> <td>4</td> </tr> <tr> <td>5.0</td> <td>5</td> </tr> <tr> <td>7.5</td> <td>12</td> </tr> <tr> <td>10.0</td> <td>12</td> </tr> <tr> <td>12.5</td> <td>1</td> </tr> <tr> <td>15.0</td> <td>1</td> </tr> <tr> <td>17.5</td> <td>1</td> </tr> </tbody> </table>	Tm Difference	Count	0.0	5	2.5	4	5.0	5	7.5	12	10.0	12	12.5	1	15.0	1	17.5	1	Variation in Tm difference among generated candidates indicates differing thermodynamic asymmetry. Favorable asymmetry improves guide strand incorporation into RISC and may increase silencing efficiency. Candidates with stronger asymmetry were ranked higher in later analyses.
Tm Difference	Count																		
0.0	5																		
2.5	4																		
5.0	5																		
7.5	12																		
10.0	12																		
12.5	1																		
15.0	1																		
17.5	1																		



Candidate ranking integrated accessibility and thermodynamic characteristics. Higher-ranked candidates demonstrated improved energetic properties and were considered more favorable for downstream evaluation. Ranking reduced the number of potential candidates requiring further analysis.



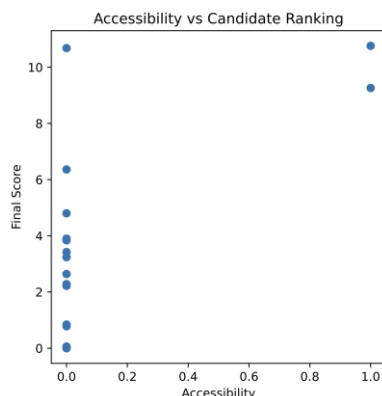
The relationship between thermodynamic properties and final candidate ranking suggests that energetic favorability contributes to candidate prioritization. Stronger thermodynamic characteristics generally corresponded with higher scores.



BLAST analysis detected sequence similarities with multiple transcript records. Although designed candidates target IL6, potential off-target interactions cannot be completely excluded. Additional specificity optimization or experimental validation would be necessary before therapeutic application.



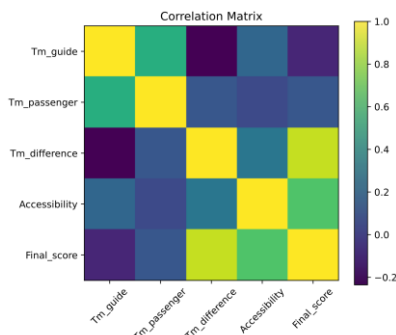
Candidate target positions were distributed across multiple transcript regions rather than concentrated within a single location. Broader distribution increases likelihood of identifying accessible and functionally favorable targets.



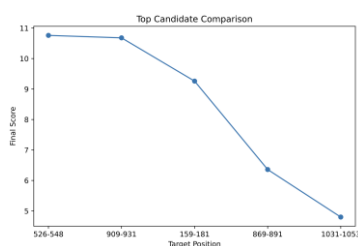
Candidates overlapping structurally accessible regions generally achieved higher ranking scores. This supports the hypothesis that accessibility contributes to improved predicted siRNA performance.



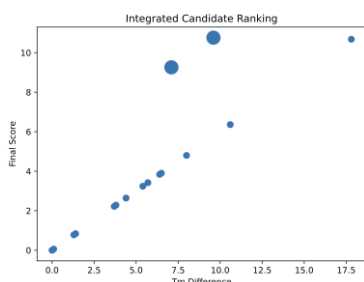
Heatmap analysis enabled simultaneous comparison of multiple candidate features including Tm values, accessibility, and ranking. Top candidates exhibited more favorable combinations of these properties relative to lower-ranked molecules.



Correlation analysis explored relationships among candidate characteristics. Positive relationships between accessibility and ranking support integration of structural information into scoring approaches. Weak correlations may indicate independent contributions of different parameters.



Comparison of highest-ranked candidates highlighted differences in predicted suitability. This analysis supported prioritization of specific molecules for further computational evaluation.



Integrated ranking combined accessibility, thermodynamic asymmetry, and scoring information. The figure summarizes overall candidate prioritization and represents the final stage of computational selection before biological interpretation. Candidates with favorable combined properties may represent stronger therapeutic possibilities.

The computational workflow identified multiple IL6-targeting siRNA candidates exhibiting favorable accessibility, thermodynamic properties, and interaction stability. Strong duplex formation and ranking scores suggest potential effectiveness of selected molecules. However, BLAST analysis indicated possible off-target interactions, emphasizing the importance of specificity evaluation. The results collectively support the feasibility of IL6-targeting siRNA therapeutics while highlighting the need for experimental validation and advanced stability studies.

Deviations from Proposal:

Original Proposal	Implemented Approach	Reason for Change	Effect on Project
Full molecular dynamics (MD) simulation of siRNA–mRNA complex	Preliminary stability evaluation using energetic interpretation and thermodynamic properties	Full atomistic MD requires validated 3D structures, force-field parametrization, extensive simulation time, and computational resources beyond project scope	Allowed approximate stability assessment while maintaining computational feasibility
General stability analysis	Integrated thermodynamic ranking using accessibility + Tm difference + energetic properties	Combined scoring provided a more interpretable method for prioritizing candidates	Improved candidate comparison and ranking
Broad specificity evaluation	BLAST+ off-target transcript screening	BLAST provides reproducible and established sequence similarity assessment	Increased confidence in specificity evaluation and identified potential off-target effects
Conceptual therapeutic discussion only	Expanded analysis including IL4–IL13 axis and cytokine network interpretation	Broader biological interpretation improved translational relevance of findings	Strengthened connection between computational outputs and disease biology
General delivery strategy discussion	Specific localized delivery approaches considered (intranasal delivery, lipid nanoparticles, polymer carriers)	Therapeutic success depends strongly on delivery efficiency and reduction of systemic effects	Improved practical relevance of proposed therapeutic application
Limited visualization planning	Additional quantitative visualizations generated including ranking plots, accessibility analysis, heatmaps, and correlation matrices	Visual interpretation improved understanding of candidate behavior and supported conclusions	Enhanced presentation and interpretation of computational results

The most significant deviation involved replacement of full molecular dynamics simulations with stability approximation methods. Although molecular dynamics would provide deeper structural insights, implementation would require experimentally validated three-dimensional duplex structures and extensive computational resources. Therefore, thermodynamic interpretation and energetic evaluation were adopted as practical alternatives.

Additionally, biological interpretation was expanded beyond the original proposal through integration of IL4 and IL13 inflammatory pathways. This modification improved understanding of how IL6 suppression may influence broader cytokine networks associated with allergic inflammation.

The inclusion of BLAST-based specificity screening and expanded visual analysis also strengthened reproducibility and interpretability of the workflow.

Overall, deviations from the original proposal were introduced to improve feasibility, maintain methodological consistency, and produce a more complete computational pipeline.

Limitations:

The present study is entirely computational and therefore lacks experimental validation of predicted siRNA efficacy. Stability assessment was based on thermodynamic interpretation rather than full molecular dynamics simulations, and off-target analysis relied primarily on sequence similarity screening. In addition, cytokine interactions were interpreted from existing literature and were not experimentally evaluated. Consequently, findings should be considered preliminary predictions requiring further biological validation.

Future Work:

Future studies may incorporate experimental validation of selected siRNA candidates through in vitro or in vivo models to assess actual gene silencing efficiency and specificity. Advanced molecular dynamics simulations, improved off-target prediction methods, and optimized delivery systems such as lipid nanoparticles or localized intranasal formulations may further strengthen therapeutic evaluation. Additional investigation of IL4–IL13–IL6 interactions could also improve understanding of broader cytokine regulation in allergic inflammation.

Conclusion:

This study developed a computational workflow for the design and evaluation of IL6-targeting siRNA candidates within the context of allergic inflammation and cytokine regulation. The workflow integrated transcript retrieval, siRNA generation, RNA secondary structure prediction, duplex interaction analysis, thermodynamic ranking, specificity screening, and biological interpretation.

Results identified candidate siRNAs exhibiting favorable accessibility, predicted interaction stability, and acceptable thermodynamic properties. Off-target analysis suggested that specificity remains an important consideration, highlighting the need for further optimization before therapeutic application. Integration of IL4 and IL13 inflammatory pathways additionally emphasized the complexity of cytokine networks and the importance of evaluating broader biological effects of IL6 suppression.

Although experimental validation was beyond the scope of this study, findings demonstrate the potential usefulness of computational bioinformatics approaches for early-stage RNA therapeutic development. Overall, the project supports the feasibility of IL6-targeting siRNA strategies while illustrating how structural bioinformatics tools may contribute to therapeutic candidate prioritization.

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